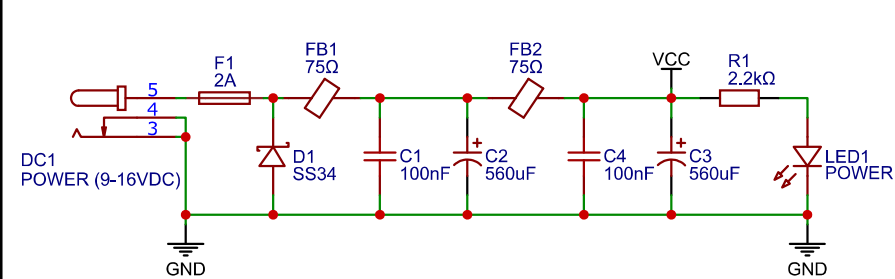
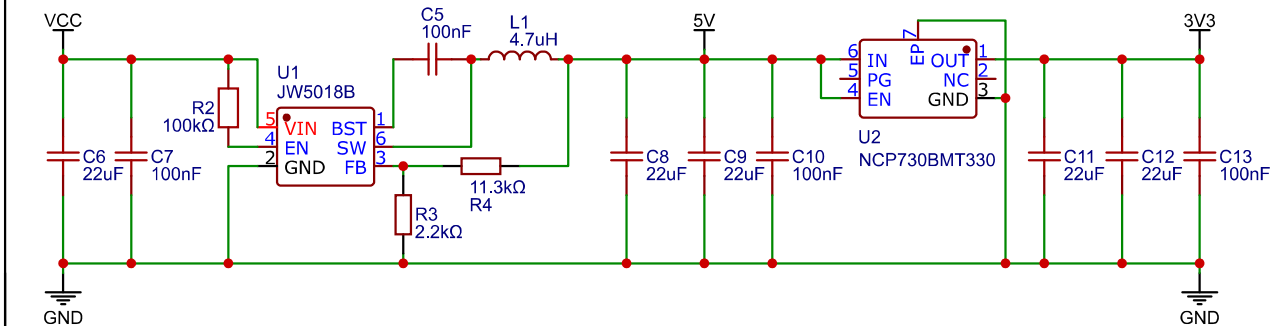


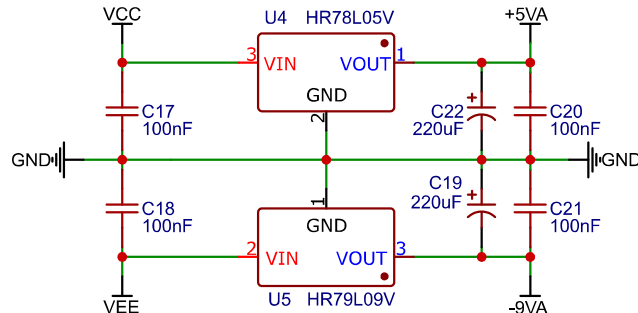
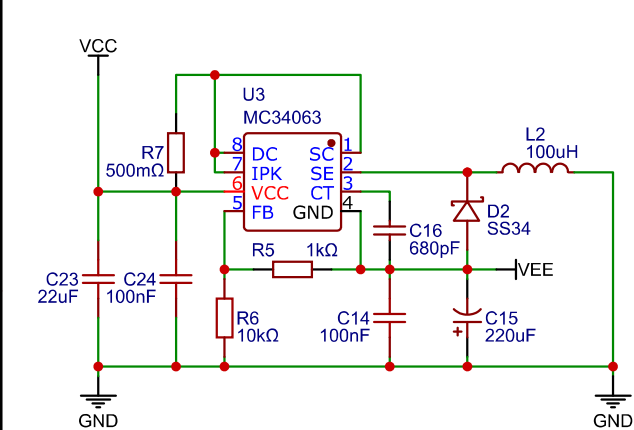
POWER INPUT



DIGITAL SUPPLY

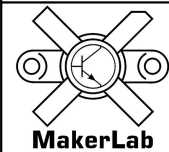


ANALOG SUPPLY



TEST POINTS

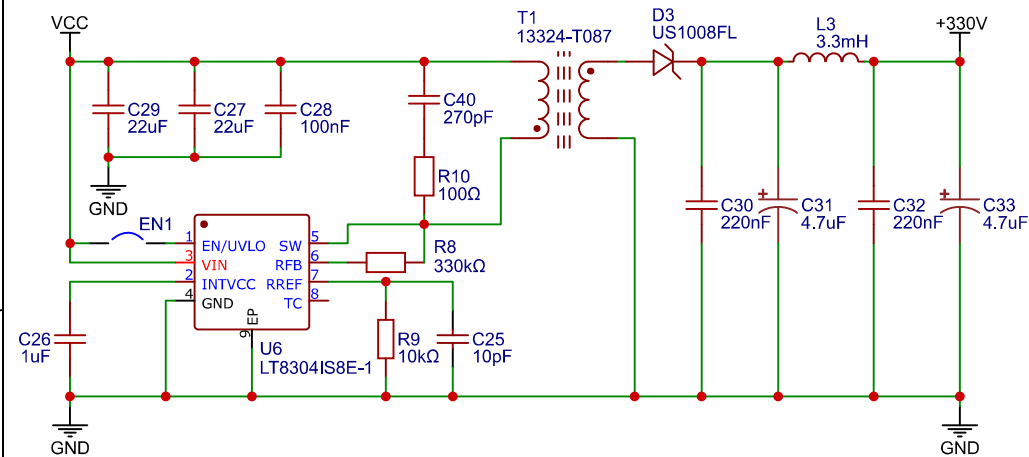
TP1 VCC	This is your input supply voltage. Make sure it fits in 9-16VDC range. Otherwise the board will function incorrectly or can be damaged!
TP2 5V	This is 5VDC output only for 3V3 LDO for ESP32 and DS3231 power. It's necessary for keeping ESP32 DAC noise floor low.
TP3 3V3	This is 3V3 LDO out to power ESP32 and DS3231 RTC. It's necessary for keeping ESP32 DAC noise floor low.
TP4 +5VA	This is +5V analog filtered out to power analog stuff.
TP5 -9VA	This is -9V analog filtered out to power analog stuff.



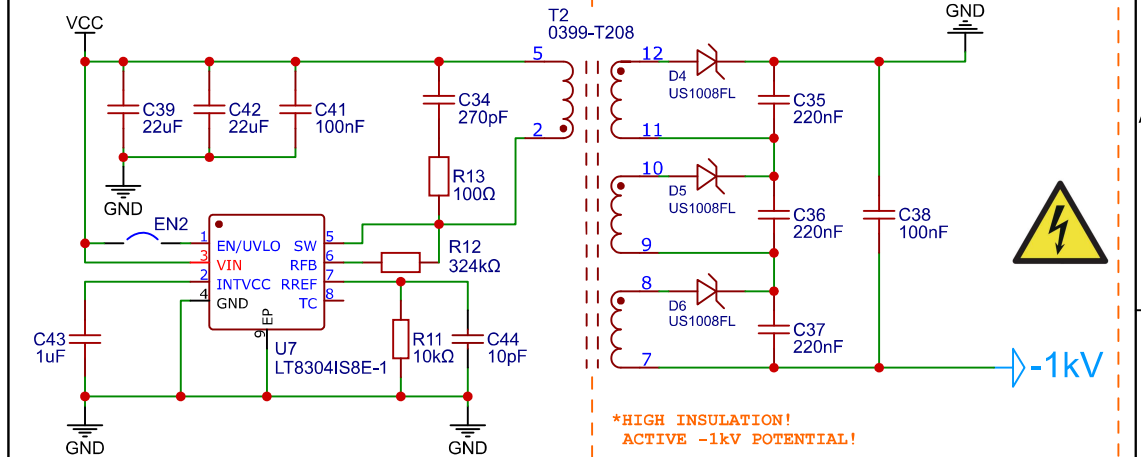
ScopeClock ECC-0755

TITLE:	Power Input	REV: 1.0
	Company: MakerLab	Sheet: 1/1
	Date: 2026-06-09	Drawn By: MakerLab

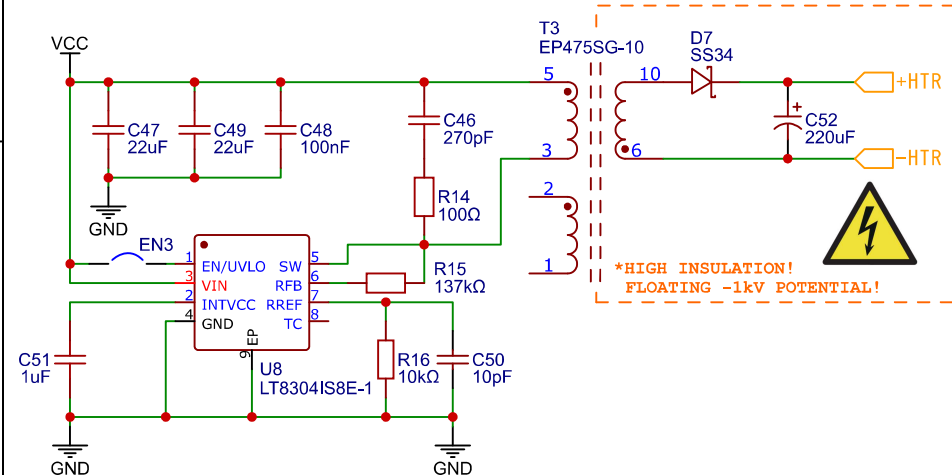
DEFLECTION SUPPLY



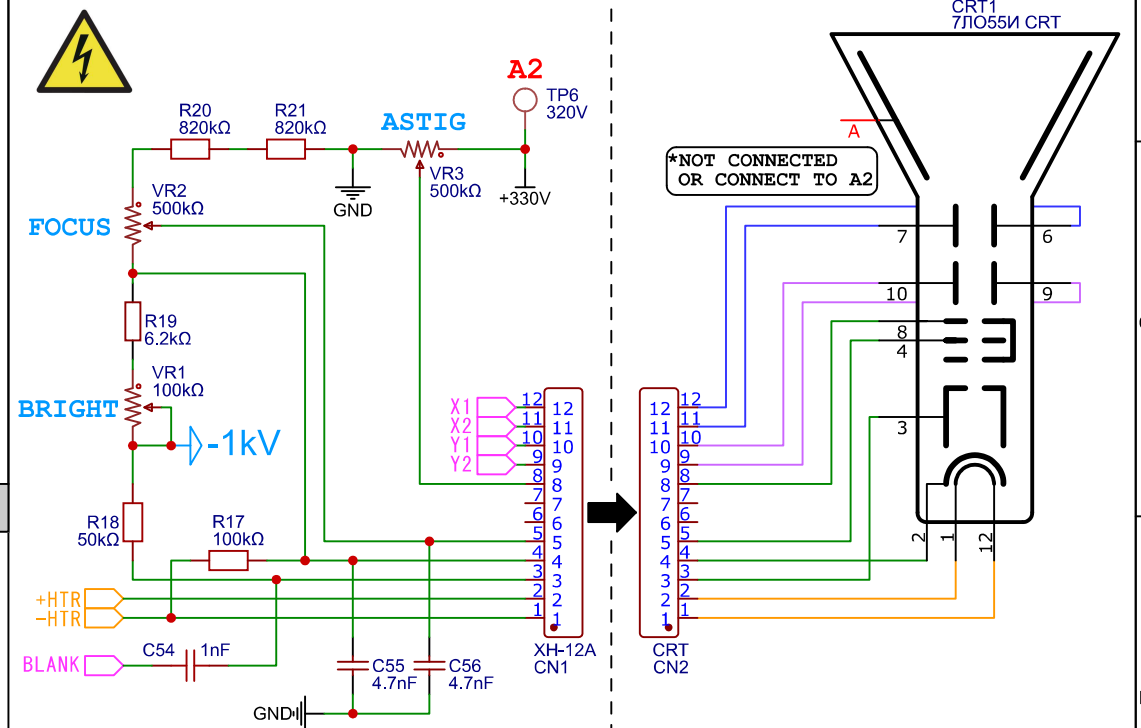
CATHODE SUPPLY



HEATER SUPPLY



CRT CONNECTION



NOTES

- Solder all the jumpers:
- EN1: Check for 330V anode voltage.
 - EN2: Check for -1kV cathode voltage.
 - EN3: Check for 6.3V heater voltage.



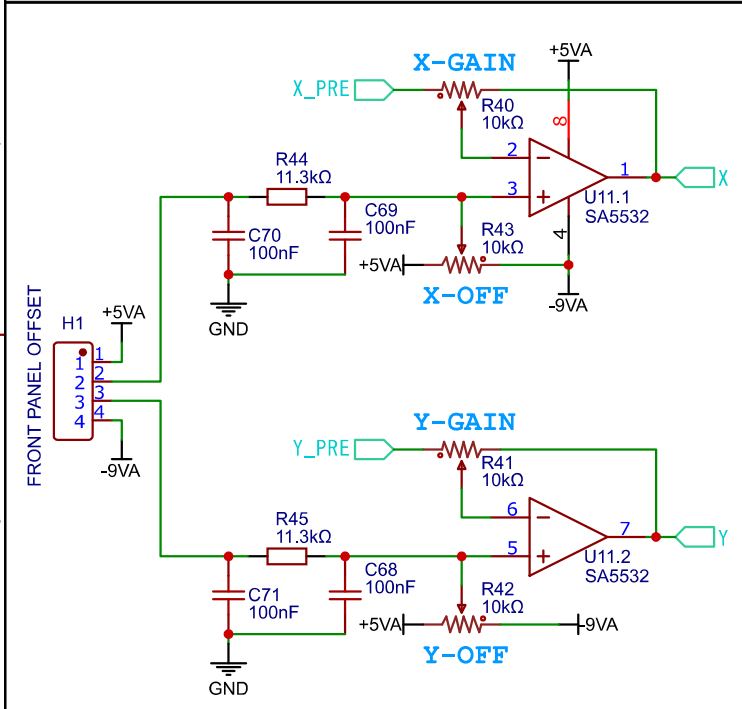
HIGH VOLTAGE!
ACTIVE 1.5kV DC



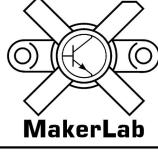
ScopeClock ECC-0755

TITLE: Tube Supply		REV: 1.0
Company: MakerLab		Sheet: 1/1
Date: 2026-06-09	Drawn By: MakerLab	

XY PREAMP



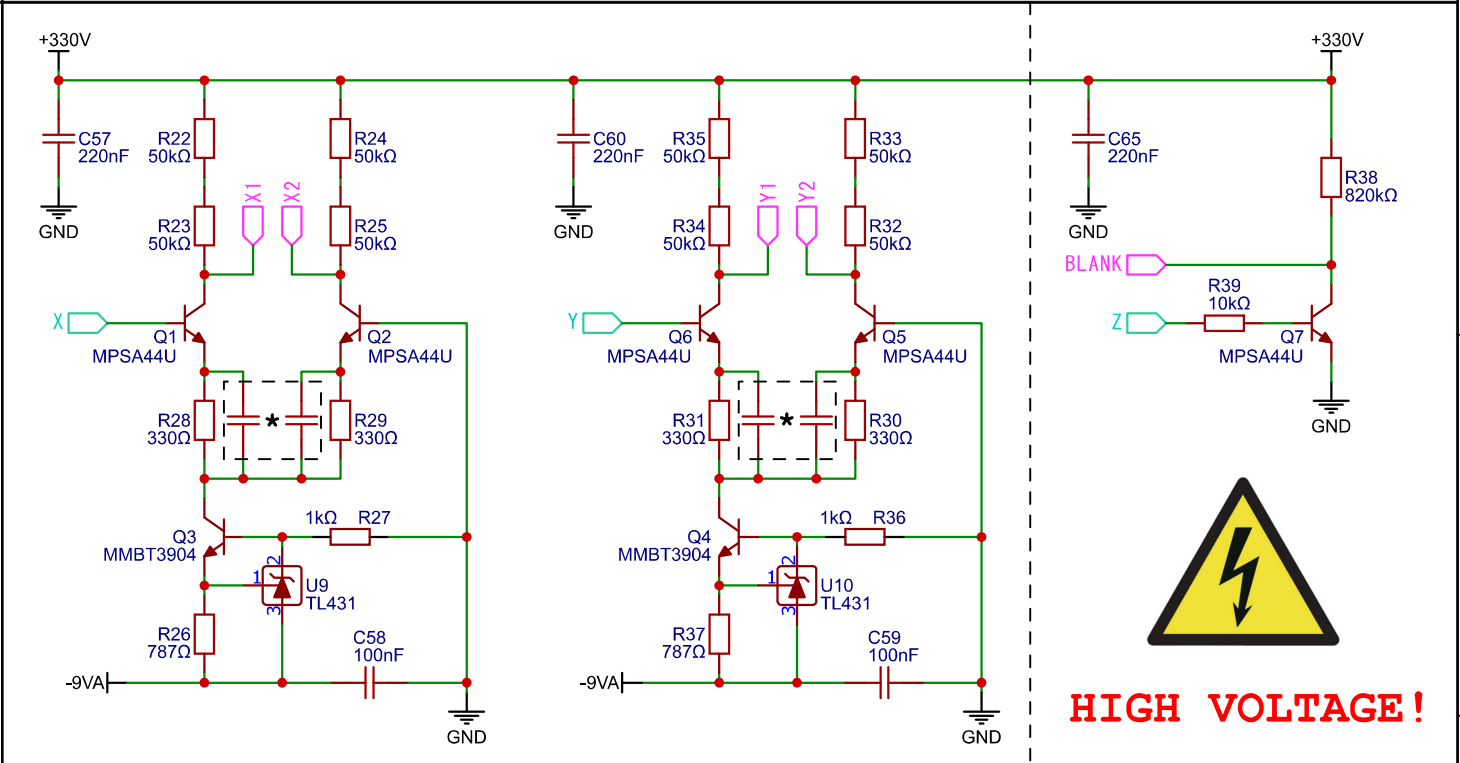
You need to properly set X and Y gain to fit image on the screen. Also it's needed to precise tune image offset.
(Set X-OFF and Y-OFF pots to 0V)



ScopeClock ECC-0755

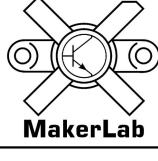
TITLE: Deflection		REV: 1.0
Company: MakerLab		Sheet: 1/1
Date: 2026-06-09	Drawn By: MakerLab	

XYZ DEFLECTION AND BLANKING AMPLIFIER



If image is mirrored by X or Y just swap CRT deflection plates pins.
Set X-GAIN and Y-GAIN to fit image on the screen. Then you don't need to touch it untill CRT swap.
Set X-OFF and Y-OFF pots to 0V measured at H1. Then you can fine offset image by front panel potentiometers.

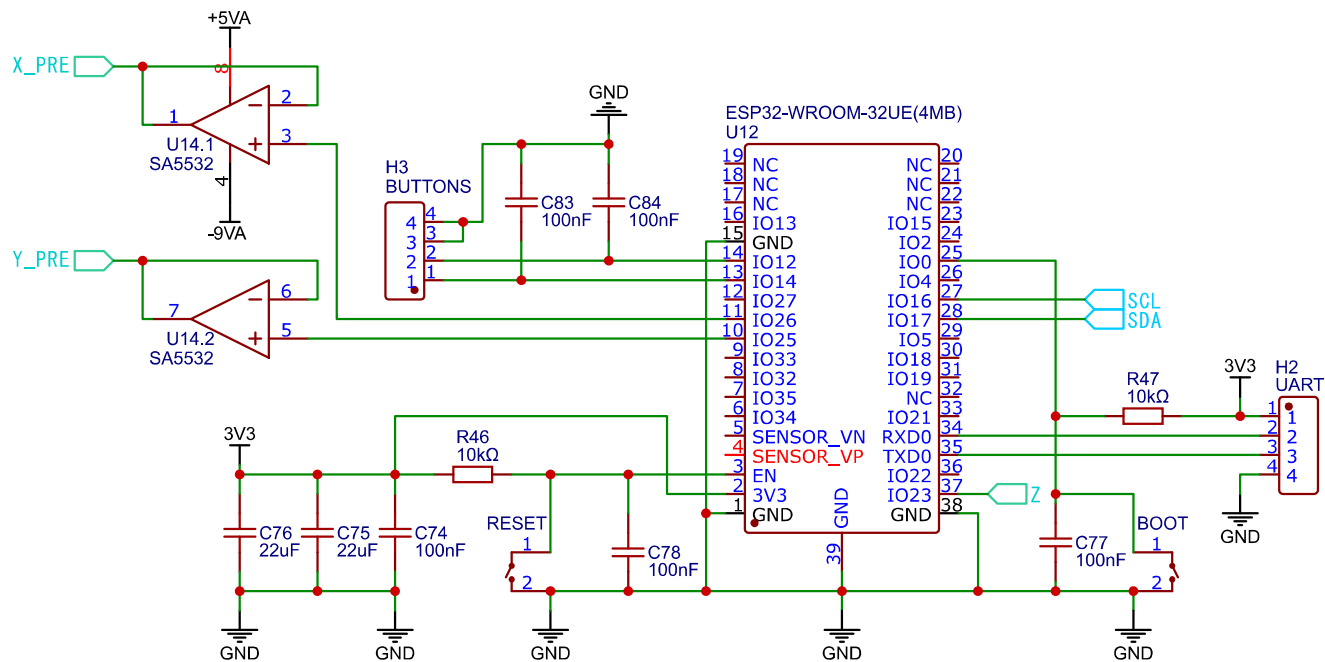
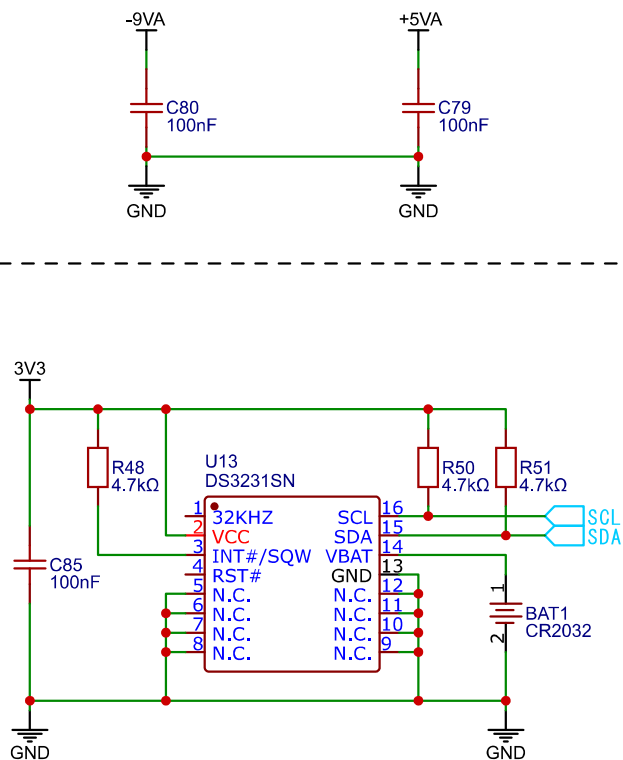
* Compensation capacitors must be tweaked with individual values for each tube.



ScopeClock ECC-0755

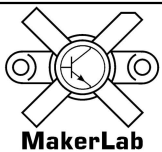
TITLE: Deflection		REV: 1.0
Company: MakerLab		Sheet: 1/1
Date: 2026-06-09	Drawn By: MakerLab	

MCU CLOCK



RTC uses standart CR2032 3V battery.

To flash firmware to ESP32 connect it to UART and set flash speed to 115200 baud. Right before flashing hold BOOT and while holding it also push RESET. While still holding BOOT release RESET. ESP32 is now in boot mode and you can release BOOT too. Now you can flash firmware. After that push RESET. Solder front panel Button 1 and Button 2 to "BUTTONS" H3.



ScopeClock ECC-0755

TITLE:

MCU Clock

REV: 1.0

Company: MakerLab

Sheet: 1/1

Date: 2026-06-09 Drawn By: MakerLab